8.1.2.1 Nosé-Poincaré Dynamics: A Symplectic Approach

As we have seen in earlier chapters, there is a strong foundation for using symplectic integrators, for example the existence of perturbative backward error expansions which are relatively easily calculated, it is desirable to work within this class. On the other hand, as we have mentioned, Nosé-Hoover dynamics is not a Hamiltonian system. The Nosé-Poincaré method resolves this issue by providing a Hamiltonian-based extension of the physical system whose projected trajectories are actually equivalent to Nosé-Hoover trajectories, in the absence of numerical errors. This extended Hamiltonian system can then be discretized using a symplectic method.

Nosé's original proposal involved using the dynamics of the Hamiltonian

$$H_{\mathrm{N}} = \frac{1}{2s^2}\tilde{\boldsymbol{p}}^T\boldsymbol{M}^{-1}\tilde{\boldsymbol{p}} + \frac{\pi^2}{2\mu} + U(\boldsymbol{q}) + (\mathrm{N_d}+1)\mathrm{k_B T}\ln s$$

to define an extension which can be used for canonical sampling. The dynamics for constant Nosé extended energy, when reduced to the $(\boldsymbol{q}, \boldsymbol{p} = \tilde{\boldsymbol{p}}/s)$ plane will preserve the canonical measure ρ_β. However, the scaling of kinetic energy induces a time transformation, which is inconvenient for molecular modelling purposes. The Nosé-Poincaré method introduces an explicit unraveling of the time transformation. One works with the modified Nosé-Poincaré (NP) Hamiltonian [41, 102]:

$$H_{\mathrm{NP}} = s[H_{\mathrm{N}} - \mathrm{E_N}] = s\left[\frac{1}{2s^2}\tilde{\boldsymbol{p}}^T\boldsymbol{M}^{-1}\tilde{\boldsymbol{p}} + \frac{\pi^2}{2\mu} + U(\boldsymbol{q}) + \mathrm{N_d k_B T}\ln s - \mathrm{E_N}\right], \tag{8.9}$$

where $\mathrm{E_N}$ is a constant equal to the initial Nosé energy of the system. Trajectories of this system unfold on a timescale which is in some sense commensurate with that of Hamiltonian dynamics. The NP Hamiltonian system can be treated with symplectic integrators, conferring a possible stability advantage. Symplectic integrators for this system are compared in [41].

The equations of motion associated to (8.9) are

$$\frac{\mathrm{d}}{\mathrm{d}t}\boldsymbol{q} = \boldsymbol{M}^{-1}\tilde{\boldsymbol{p}}/s, \tag{8.10}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}\tilde{\boldsymbol{p}} = s\boldsymbol{F}, \tag{8.11}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}s = sp_s/\mu, \tag{8.12}$$

$$\frac{\mathrm{d}}{\mathrm{d}t}p_s = \frac{\tilde{\boldsymbol{p}}^T\boldsymbol{M}^{-1}\tilde{\boldsymbol{p}}}{s^2} - \mathrm{N}_d\mathrm{k_B T} - \Delta H_{\mathrm{N}}, \tag{8.13}$$

where $\Delta H_{\mathrm{N}} = H_{\mathrm{N}} - H_{\mathrm{N}}^0$ (H_{N}^0 represents the value of the Nosé Hamiltonian at the initial point).